Machine Learning in Engineering Science
Week 9: Mechanisms, Controlled Experiments and Defense

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1 Week 9: mechanisms before remedies

Monday, 2 November 2026, 14:00–17:00. This week assumes students can independently explain train/validation/test and MAE/RMSE. Basic metric review takes at most five minutes. Three experiments occupy 100 minutes of practice and discussion; mechanisms and summary take 60 minutes, with ten minutes of break and ten of operating buffer. Three 30-page slide modules distinguish live core pages from untimed reference pages. These exercises are formative classwork. The midterm consists of **written and oral examinations only; there is no midterm project**. The stock forecasting Homework remains a separate designated Homework.

1.1 Five judgments

For an unfamiliar small engineering problem, independently state and defend the target, data split, baseline, evaluation and limitations. Running an AI-written notebook is insufficient: the student must explain which assumption a code modification changes, what remains fixed and which observation supports the conclusion.

1.2 One transparent comparison

Let X_0, X_1, X_2 be independent standard-normal inputs and let

$$Y = f(X) + \epsilon, \quad f(X) = 2X_0 + X_1^2, \quad \epsilon \sim \mathcal{N}(0, 0.2^2), \quad \epsilon \perp X.$$

All inputs are dimensionless; the response has synthetic response units. These are neither measured equipment data nor stock observations. X_2 is a negative control. Knowing the generator makes the mechanism identifiable in this constructed experiment, without establishing physical causation.

Generate 500 rows with seed 2026. Split 350 train and 150 held rows using seed 2026; split held rows into 75 validation and 75 test using seed 2027. Sort IDs within splits and assert disjointness. Candidates are declared before test inspection: an affine Ridge pipeline and a pipeline that appends X_1^2 before the same train-fitted standardization and Ridge ($\alpha = 1$, SVD solver). Choose by validation MAE, ties favoring affine. Freeze the choice before the predeclared two-model test diagnosis. Test diagnosis does not reopen selection.

2 Experiment A: model form, noise and repeated fitting

2.1 Derive the population affine projection

The population, unregularized least-squares affine approximation is

$$g_*(X) = 1 + 2X_0.$$

To see why, $E[X_1^2] = 1$, $E[X_1^3] = 0$ by symmetry, and independence eliminates cross-covariances. The intercept absorbs the mean square; a linear coefficient on X_1 cannot represent its even curvature. This is not an assertion that finite, regularized Ridge equals the population projection exactly.

Declare residual sign $e = \widehat{y} - y$. For this population predictor,

$$e = 1 - X_1^2 - \epsilon, \qquad E[e \mid X_1] = 1 - X_1^2.$$

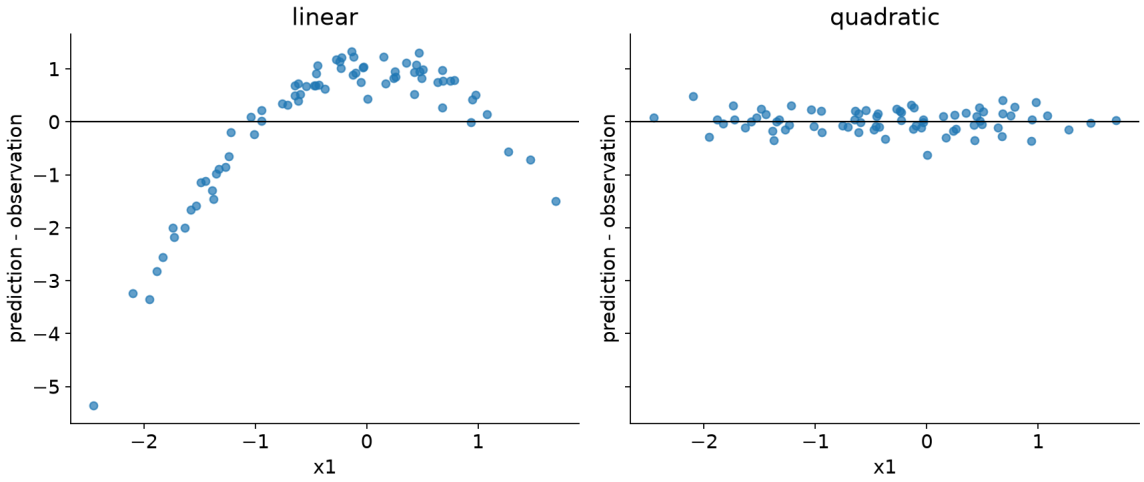
At $X_1 = 0$ the conditional mean is $+1$; at $X_1 = \pm 2$ it is -3. Positive means overprediction, negative underprediction. Finite fitted residuals also contain coefficient-estimation error and observation noise. Compare candidates on identical test rows and shared axes before opening a zoom.

2.2 Observation noise is not missing shape

For a standard normal variable, $E[X_1^4] = 3$, hence $\mathrm{Var}(X_1^2) = 3 - 1 = 2$. The independent components give

$$E[e^2] = \mathrm{Var}(X_1^2) + \mathrm{Var}(\epsilon) = 2 + 0.04 = 2.04.$$

The exact quadratic oracle has expected MSE 0.04. These population expectations are not exact finite fitted test scores. More observations can stabilize an affine fit; they cannot add a missing square to its feature family.



Actual notebook diagnosis on the same 75 test rows and shared scales. Residual sign is prediction minus observation.

2.3 Measure training-sample variation by retraining

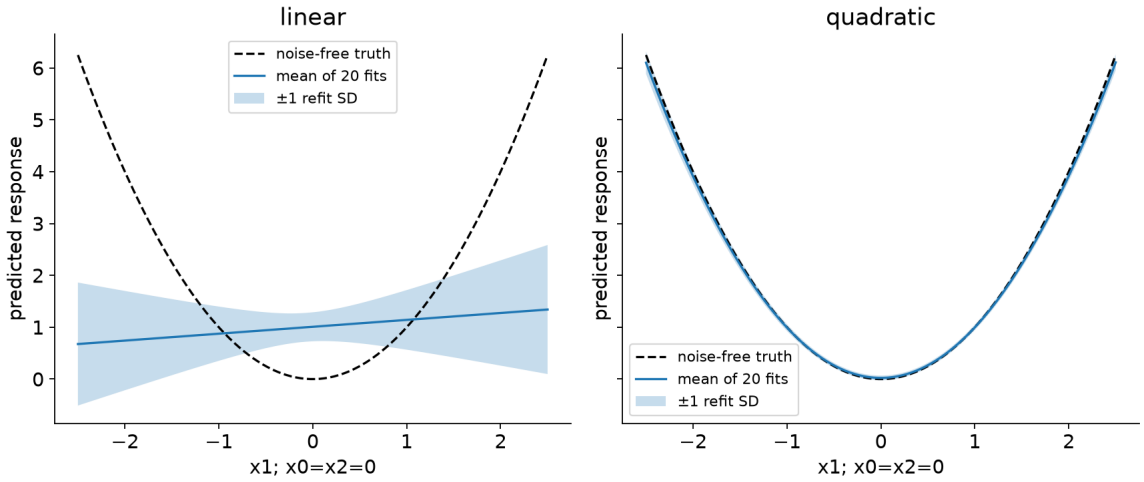
For each $n \in \{40, 160, 640\}$, fit each family on 20 independent training draws. Within a repetition, pair the model families and use prefix training rows across sample sizes. Predict the same 240-point, noise-free probe. A residual scatter from one fitted model cannot identify variation across training samples. Training noise still affects each fitted model; noise-free probe targets remove fresh label noise from this particular comparison.

Write predictions as $\widehat{f}_r(x)$, $r = 1, \ldots, R$, with $\bar{f}(x) = R^{-1} \sum_r \widehat{f}_r(x)$. At each probe point, subtract and add $\bar{f}(x)$; the cross term sums to zero:

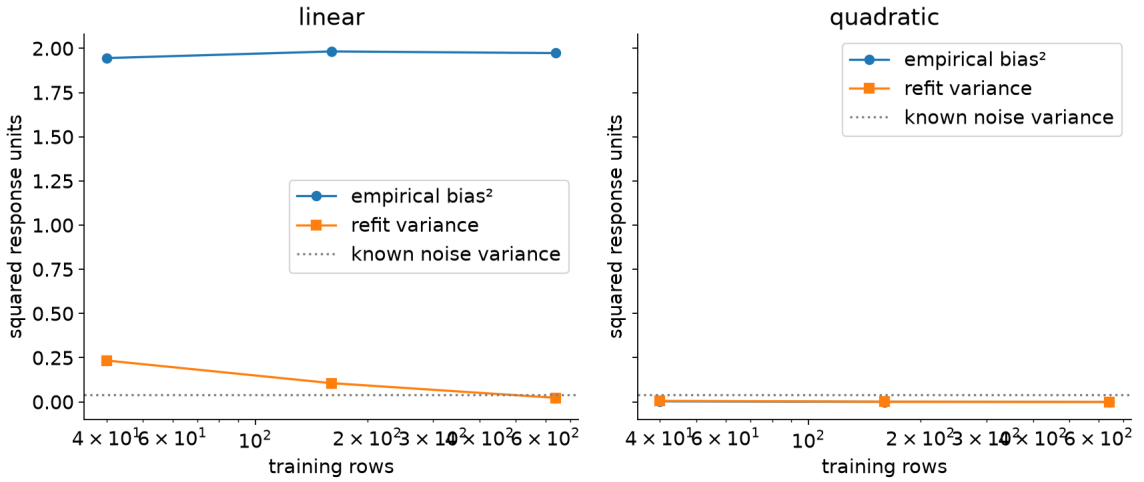
$$\frac{1}{R} \sum_r (\widehat{f}_r - f)^2 = (\bar{f} - f)^2 + \frac{1}{R} \sum_r (\widehat{f}_r - \bar{f})^2.$$

Average over the fixed probe points. This is an exact finite-array identity when variance uses denominator R (`ddof=0`). The empirical mean is not the exact infinite-training-population mean.

Adding the known 0.04 gives the noise-averaged expected label-error component for this ensemble, not an equality to one noisy held-out test score. Twenty repetitions support a classroom estimate, not a universal monotonic learning curve. Shaded ±1 refit standard deviations are not predictive intervals.



Actual independently refitted models, 40 training rows per fit. The mean curve and spread answer different questions.



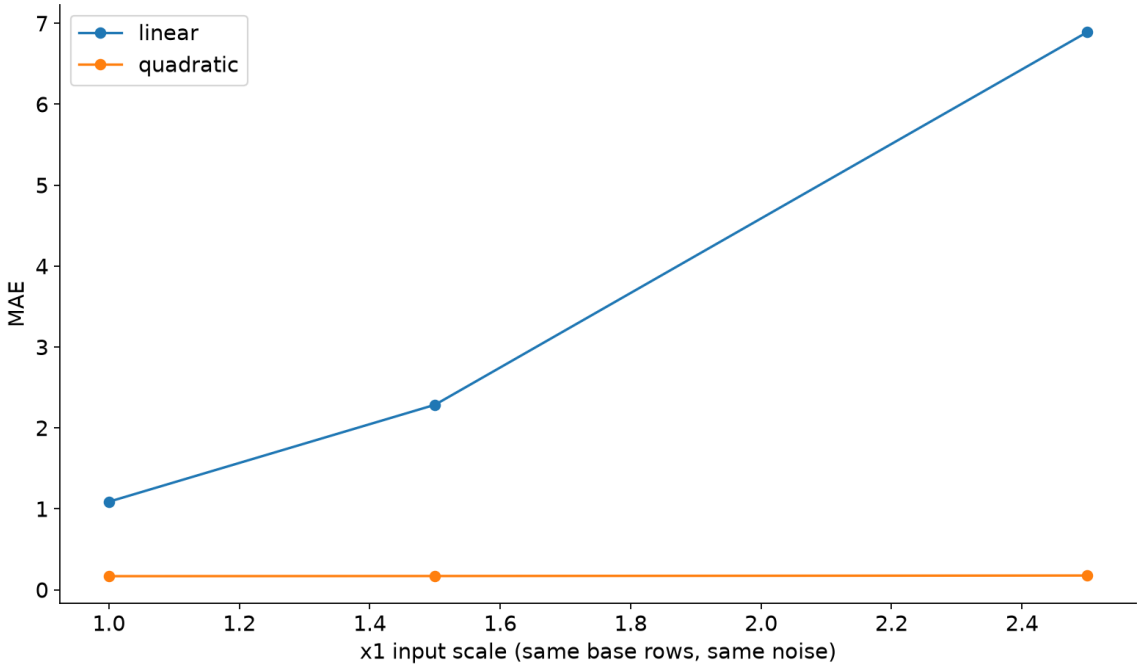
Actual finite empirical decomposition on a fixed noise-free probe. Both panels share the vertical scale.

Practice, 35 minutes. Predict which component should change with more rows, run the paired refits, inspect shared-scale residuals and bands, then cite an observed result and defend a mechanism. State one limitation. Do not copy a worked response before predicting.

3 Experiment B: inputs versus target relationships

3.1 A paired input-scale stress test

Draw 150 base rows with seed 2126. Multiply only X_1 by 1, 1.5, or 2.5. Keep other inputs, base row identity, noise and the target rule fixed; recompute labels from the same rule. Keep model parameters frozen. Plot MAE against scale without refitting or reselection. Gaussian support remains unbounded: probability mass and finite observed coverage change, not mathematical support. An input shift need not worsen every model; risk depends on which cases become more common and their conditional errors.



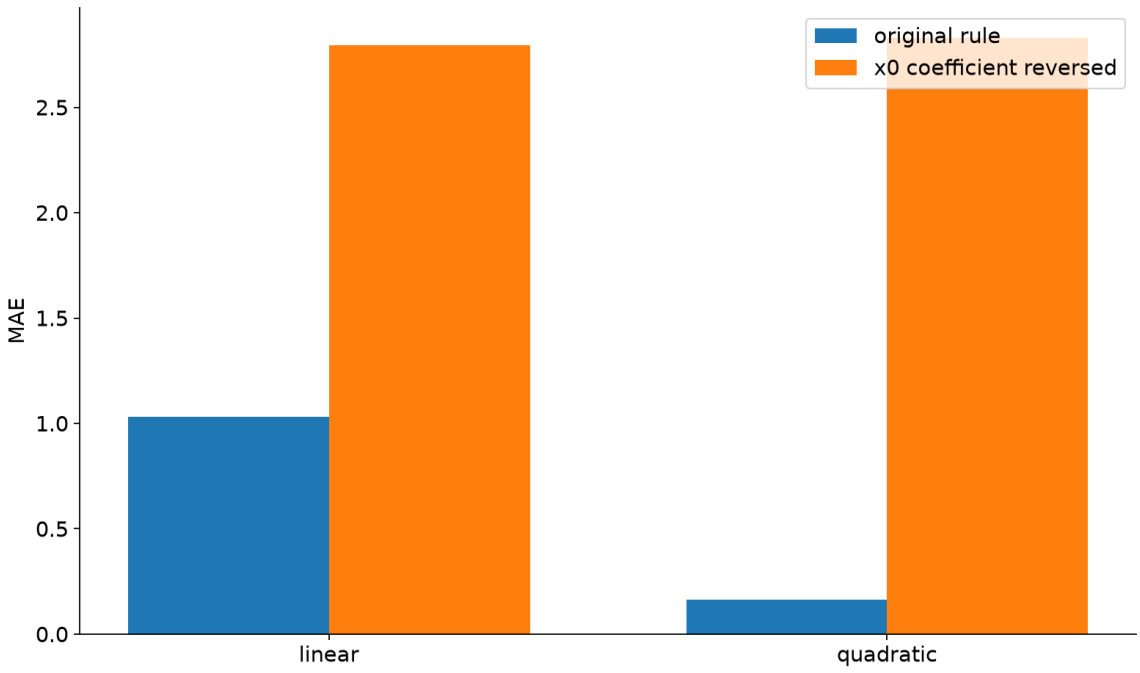
Actual notebook results on paired synthetic stress rows. These outcomes are diagnostics, not a model-selection round.

3.2 A changed target relationship with identical inputs

On the exact original test inputs and noise, replace the coefficient $+2$ on X_0 with -2. Frozen predictions stay unchanged. Since $y_{\text{new}} = y_{\text{old}} - 4X_0$,

$$e_{\text{new}} = e_{\text{old}} + 4X_0.$$

Input monitoring alone cannot reveal this label change. Timestamped labels, residuals, schema/unit checks and domain context are needed. Several changes may coexist in practice; these two synthetic interventions are controlled examples, not exhaustive causal diagnosis.



Identical test inputs, paired noise, frozen models; only the target relationship changes.

3.3 Why a single average is insufficient

For a paper illustration with group losses 1 and 10 and difficult-group fraction q, aggregate MAE is $(1-q)1+q10 = 1+9q$. It changes from 1.9 at $q = .10$ to 3.25 at $q = .25$ without changing either group loss. These illustrative counts are not the fitted-model test counts. Report group definitions, counts and conditional losses together.

Practice, 25 minutes. Predict both interventions before execution. For each, identify the changed factor, controlled quantities, measured outcome and claim boundary. Then take the ten-minute break; five minutes of operating buffer are reserved in this hour.

4 Experiment C: independently modify and defend

4.1 A real code change

Edit the provided X-only feature function. Preserve original columns first; append a deterministic transformation. Never pass labels, row IDs or split membership. Validate shape, finiteness, determinism and absence of input mutation. The starter identity intentionally remains a runnable draft. Filling response fields without changing code does not implement an experiment.

State the hypothesis, expected outcome and held-fixed conditions before fitting. Use the original training rows, scaler protocol and Ridge settings. Inspect validation, then record the source and plan digest before materializing a new 240-row independent synthetic audit sample. Preserve the selected baseline. Evaluate both on identical audit rows. Repeated tuning against audit outcomes is exploratory and must be disclosed; a digest records bytes, not honesty or correctness.

A negative result is acceptable. For example, a different symmetry in a proposed basis may not repair the missing even relationship. The student's task is to connect an actual code change to a measured observation and a bounded explanation, not to guarantee a lower score.

4.2 Evidence and replay scope

Export exact CSV, IDs, saved feature source, frozen declaration, baseline prediction vectors, metrics, figures, responses and environment versions. The edited notebook must be saved separately. The CSV readback refits the already selected reference in the same installed environment and checks aligned predictions. This is same-environment consistency, not independent environment reproduction or scientific validation. Week 10 extends portability.

The notebook's *ready for peer review* status only checks that code differs from the identity and required fields are present. It is not an automated grade, correctness claim or certificate of independent competence.

Practice, 40 minutes. Predict, edit, check, freeze, audit once, inspect the export and explain to a partner. A final five-minute operating buffer supports recovery. Defend target, split, baseline, evaluation and limitations without reading generated prose.

5 Summary and transfer

Missing representation differs from refit variability and observation noise. Input shift differs from a changed target relationship. Preserve a baseline and choose evidence that matches the question. For a new engineering problem with stable input histograms but worse labeled error, propose competing explanations and the observations needed to distinguish them. Written and oral examination questions may test these independent judgments; no midterm project is required.

6 Primary references

- scikit-learn, bias–variance example: https://scikit-learn.org/stable/auto_examples/ensemble/plot_bias_variance.html.
- scikit-learn, common pitfalls and leakage: https://scikit-learn.org/stable/common_pitfalls.html.
- Google, monitoring ML systems: https://developers.google.com/machine-learning/crash-course/production-ml-systems/monitoring.